

Time resolution of wind speed data introduces errors in wind power density assessment

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ABSTRACT

Wind energy is critical in the global shift towards renewable energy sources, necessitating accurate potential assessments for informed decision-making and strategic development in this transition. Despite the widespread use of wind speed data with varying time resolutions for estimating wind energy, the influence of this variability on the accuracy of wind energy assessment remains uncertain. In this study, we investigate the impact of wind speed data's time resolution on wind power density (WPD) assessment, utilizing high-frequency, *in-situ* observations collected from eight anemometer towers in China. Through analysis of wind speed data at nine resolutions, ranging from 10-minute to monthly intervals, we uncover significant errors introduced by coarser time resolutions in WPD calculations. Specifically, it reveals a systematic underestimation of WPD with decreasing time resolution, highlighting the importance of time resolution in wind energy evaluation. While hourly wind speed data yield acceptable WPD errors, daily and monthly resolutions lead to substantial underestimation due to the smoothing effect of fluctuations. We also delve into the underlying mechanisms contributing to these errors, including variations in the power law exponent and changes in the Weibull distribution shape factor with time resolution. Furthermore, we propose a novel method to mitigate WPD calculation errors using coarse time resolution data by regressing Weibull distribution factors. The findings offer valuable insights into wind speed data selection and accurate wind energy assessment, with implications for renewable energy planning and policy-making in the context of the global energy transition.

1. Introduction

Wind energy plays a significant role in meeting the 1.5 °C climate target. In 2023, it achieved a remarkable milestone by adding over 100 GW of new capacity globally, a nearly 60 % increase that sets a new record [1]. The trajectory continues to be promising, with the global wind energy market projected to grow by an average of 15 % annually between 2023 and 2027 [2], highlighting the importance of accurately assessing wind energy to navigate its evolving landscape. Generally, high-frequency wind speed data offers greater precision in quantifying wind energy potential. However, because of the high costs and difficulties of obtaining high-frequency data, many studies resort to using data with limited frequency. These studies utilize wind speed data at various resolutions, ranging from minute-level [3,4], to hourly [5], daily

[6,7] or even coarser time resolution. There are also studies that attempt to standardize data resolution by averaging fine-resolution data to produce coarser time resolutions [5,8]. However, the impact of varied time resolutions on wind energy assessment errors remains unclear.

Numerous studies have utilized wind speed data with varying time resolutions to evaluate wind energy by calculating wind power density (WPD), a quantitative measure of wind energy corresponding to the power available per square meter of swept area of a turbine [3–5,7,9–11]. Typically, studies utilize the readily available 10 m wind speed data, extrapolating it to hub-height wind speeds using the power law that effectively describes the near-surface wind speed profile [12]. Subsequently, they apply the probability density distribution such as the widely used two-factor Weibull distribution that is widely applied in most of relevant research to calculate WPD through downscaling

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[13–15]. Regarding wind speed extrapolation, numerous studies have analyzed upper-level wind speed data from various regions worldwide to examine the temporal and spatial variations in the power law exponent and assess the accuracy of wind speed extrapolation [16–19]. Some research specifically focuses on the differences in power law exponents derived from wind speeds at different heights [20,21]. In terms of Weibull regression, many studies compare and analyze the performance of the Weibull distribution against other distributions [22,23], while others investigate the temporal variations in Weibull distribution factors [24,25].

However, calculations of the two steps for wind power density assessments are sensitive to the time resolution of the original data, a detail previously overlooked in the abovementioned studies. Coarse time resolution leads to inaccuracies in WPD values through the Weibull distribution, with studies reporting changes in Weibull factors without addressing the underlying error causes [8,26]. At the same time, current statistical methods that aim to improve wind energy assessment accuracy not only overlook core differences induced by varying time resolution, but are also based on analysis of 10 m wind speed data which may not accurately represent conditions at hub-heights [27–30]. The absence of a clear relationship between wind speed data's time resolution and wind power density (WPD), coupled with a lack of detailed exploration into the causes of errors, poses the risk of introducing systematic biases in studies evaluating wind energy potential. This limitation could significantly impact decision-making related to energy transition.

To address these gaps, we investigate the systematic rules governing the impact of wind speed data's time resolution on WPD accuracy and define acceptable error ranges for different resolutions. To achieve this objective, we analyze hub-height wind speed data across nine resolutions and identify the core factors contributing to WPD errors. Through the identification of pivotal changes, we also develop a method to mitigate errors in WPD calculation when using coarse time resolution data.

In this study, our investigation focuses on four major processes (see the flowchart in Fig. 1): 1) Investigating variations in the power law exponent α with time resolution. 2) Examining changes in Weibull distribution with time resolution. 3) Assessing theoretical WPD errors stemming from different time resolutions of wind speed data. Quantifying cumulative errors in WPD calculation from wind speed data under various time resolutions, resulting from the combined effects of the power law and Weibull distribution. 4) Proposing a method to utilize coarse-resolution data for accurate WPD estimations that are closely aligning with fine-resolution data outcomes without large computation.

2. Methods

2.1. Data source

We collect wind speed data from eight anemometer towers located across China, provided by MingYang Smart Energy (<https://www.myse.com.cn/>). Table 1 presents the detailed information on the wind speed data (see distribution map in Figure S1). The data spans from 2013 to 2020, with each tower providing a one-year time span at a 10-minute frequency. Due to the sensitivity of the instruments and the need for statistical significance, we conduct quality control to delete the data equal to the threshold in each tower in order to extract valid wind speed data. To simulate wind speed at hub-height as accurately as possible, we specifically select data recorded at the highest point of each tower. These heights range from 70 to 140 m, aligning with the current standard wind turbine heights of 50 to 150 m, which also allows for the simulation of wind power at different turbines' heights [31]. Such high-frequency data at wind turbine heights are rare and advantageous for accurate wind energy assessment [32,33]. We average the original data into hourly, 3-hour, 6-hour, 12-hour, daily, 5-day, 10-day, and monthly resolution data for subsequent analyses.

2.2. Wind speed extrapolation

When applying the power law for wind speed extrapolation, the power law exponent α is determined through linear regression using wind speed data from all heights of each tower. The power law formula is represented as formula (1) [12,34]:

$$\frac{u_1}{u_2} = \left(\frac{h_1}{h_2} \right)^\alpha \quad (1)$$

which can compute wind speed at near-surface h_1 (u_1) to wind speed at hub-height h_2 (u_2). Utilizing the least squares method for linear regression, we transform the equation into formula (2) to derive the α value:

$$\alpha = \frac{\sum_{i=2,j=1,h_1>h_2}^n \left(\ln \left(\frac{h_{1,i}}{h_{2,j}} \right) \right) \left(\ln \left(\frac{u_{1,i}}{u_{2,j}} \right) \right)}{\sum_{i=2,j=1,h_1>h_2}^n \left(\ln \left(\frac{h_{1,i}}{h_{2,j}} \right) \right)^2} \quad (2)$$

where h_1 and h_2 are the heights corresponding to Table 1, n is the number of heights of each tower, and i and j corresponding to the numbers sorted from small to large for all heights of each tower. The y-intercept of linear regression must be zero, which conforms to the

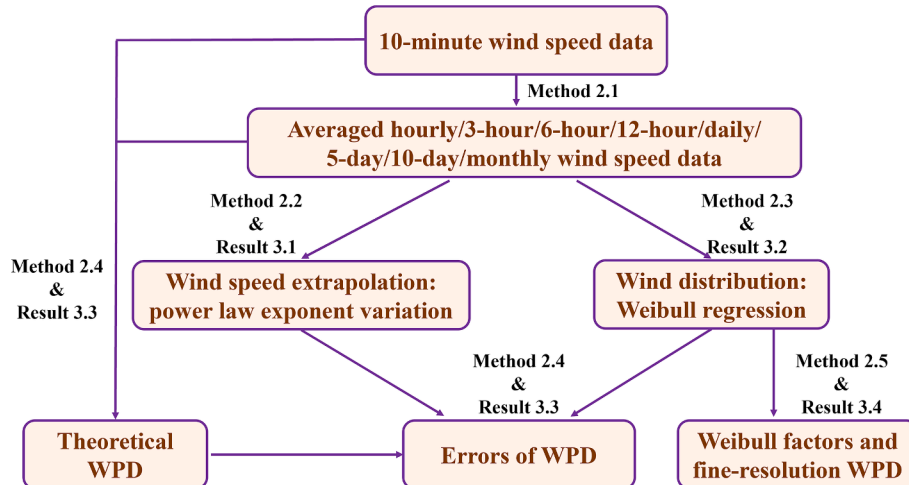


Fig. 1. Flowchart of the steps in this study. The figure shows which method and results section corresponds to each process.

Table 1
Information of wind speed data from eight anemometer towers.

Tower	Location	Terrain	Time span	Height of data	“Hub-height”	Threshold	Valid data
1	25°N 111°E	Mountain	2013.1–2013.12	10/30/50/60/70 m	70 m	0.4 m/s	96.38 %
2	43°N 88°E	Dessert	2018.1–2018.12	10/30/50/70 m	70 m	0.3 m/s	97.15 %
3	26°N 114°E	Mountain	2016.1–2016.12	10/30/50/70/80/90 m	90 m	0.4 m/s	94.33 %
4	37°N 113°E	Mountain	2019.5–2020.4	10/30/50/70/80/90 m	90 m	0.4 m/s	97.10 %
5	33°N 116°E	Plain	2018.1–2019.10	10/30/50/70/120/140 m	140 m	0.342/0.355/0.345/0.359 m/s	95.03 %
6	32°N 119°E	Plain	2019.6–2020.5	10/60/80/100/120/140 m	140 m	0.4 m/s	98.75 %
7	36°N 96°E	Dessert	2013.1–2013.12	10/30/50/70 m	70 m	0.4 m/s	96.37 %
8	32°N 120°E	Plain	2014.1–2014.12	10/30/50/70/80/90/100 m	100 m	0.4 m/s	98.47 %

physical mechanism of power law, and the reliability and significance of regression have been confirmed [19].

2.3. Weibull distribution

We conduct Weibull distribution analysis using wind speed data at the “hub-height” (Table 1) for all towers. The Weibull probability density distribution is represented by formula (3):

$$f(u) = \left(\frac{k}{c}\right) \left(\frac{u}{c}\right)^{k-1} \exp\left[-\left(\frac{u}{c}\right)^k\right] \quad (3)$$

where u is the wind speed, k is the shape factor, and c is the scale factor [35,36]. The two factors determine the width of the Weibull distribution and mean wind speed, respectively. Using all valid data, we employ the maximum likelihood method to determine the k and c factors, a commonly used approach [15,37].

2.4. Wind power density

When computing the theoretical wind power density, we include all valid hub-height wind speed data, as expressed in formula (4):

$$WPD = \frac{1}{2} \rho u^3 \quad (4)$$

where WPD represents the wind power density (W/m^2), ρ is the air density ($\rho = 1.225 kg/m^3$), and u is the averaged wind speed data at each time resolution [38].

The formula for calculating wind power density using the Weibull distribution is outlined as follows:

$$WPD = \frac{1}{2} \rho c^3 \Gamma\left(1 + \frac{3}{k}\right) \quad (5)$$

where k and c are the Weibull shape and scale factors, respectively [37,39,40]. In practical scenarios, turbines remain inactive when encountering wind speed below the cut-in and above the cut-out wind speeds, typically 3–4 m/s and 25 m/s, respectively [41]. Due to the diversity of power curves in actual situations and the successful application of formulas (4) and (5) in statistical analyses, we focus solely on the theoretical situation.

Table 2

The effect of k factor extrapolation. The table uses the k factor obtained using hourly and daily data as a reference. The extrapolated hourly k factors are obtained by regressing the daily, 2-day, 3-day, 4-day, and 5-day k factors using the power and exponential formulas respectively. k factors from regression (lines 6&9) and the coefficient of determination R^2 (lines 5&8) are the results.

Tower	1	2	3	4	5	6	7	8
k (hourly)	1.909	1.592	2.200	1.788	1.866	1.818	1.960	2.025
k (daily)	2.467	1.979	2.805	2.949	3.029	2.971	3.045	2.757
Formula	Power: $y = ax^b + 2$							
R^2	0.995	0.958	0.893	0.999	0.882	0.947	0.969	0.978
k (regression)	2.008	2.000	2.013	2.058	2.104	2a.075	2.001	2.006
Formula	Exponential: $y = ae^{bx} + 2$							
R^2	0.999	0.948	0.888	0.999	0.901	0.949	0.974	0.980
k (regression)	2.069	2.001	2.086	2.261	2.230	2.327	2.026	2.068

2.5. Regressing fine-resolution Weibull factors and WPD

To regress the k factor at coarse resolution to that at fine resolution, we utilize wind speed data at daily, 2-day, 3-day, 4-day, and 5-day resolutions (similar to [28] and employ two fundamental formulas: power and exponential, with similar shapes for regression. To enhance accuracy, we assume all k factors for 10-minute resolution data to be 2 as Rayleigh distribution [42]. Regression results are shown in Table 2, with R^2 values consistently exceeding 0.88, indicating robust performance. We extrapolate hourly k factors using the two regression curves derived from both formulas through the maximum likelihood method respectively (See Figure S2 for the process of extrapolating k factor).

3. Results and discussion

3.1. Power law exponent variation

We observe minimal variation in the annual average wind power law exponent α with respect to the time resolution, as depicted in Fig. 2, consistently remaining within a narrow range of approximately ± 0.05 . This variation is notably smaller than the temporal and spatial variations, where changes can be several times more pronounced. However, within each individual tower, α based on higher resolution displays more significant fluctuations and numerous outliers. Notably, in the 10-minute results, six of the towers display whiskers extending beyond 0.5, indicating that the vertical wind profile often stabilizes at still wind speeds over short time periods, which reflects the instantaneous condition that are rarely seen in other studies [43]. Negative values at high resolution indicate transient unstable atmospheric conditions. As the resolution coarsens, the occurrence of α outliers diminishes, which can be attributed to higher-resolution data capturing more frequent instantaneous conditions and the averaging eliminating extreme values [44]. However, achieving more stabilized α values at coarser resolutions does not necessarily translate to more accurate extrapolation. As the α value is influenced by atmospheric circulation that exhibits distinct diurnal variation, it might be overlooked at coarser resolutions [20,45]. Therefore, careful consideration must be given to selecting an appropriate time resolution for wind speed extrapolation. Furthermore, in terms of spatiotemporal variation, our study's α values for the eight towers closely align with findings in other literature. When the

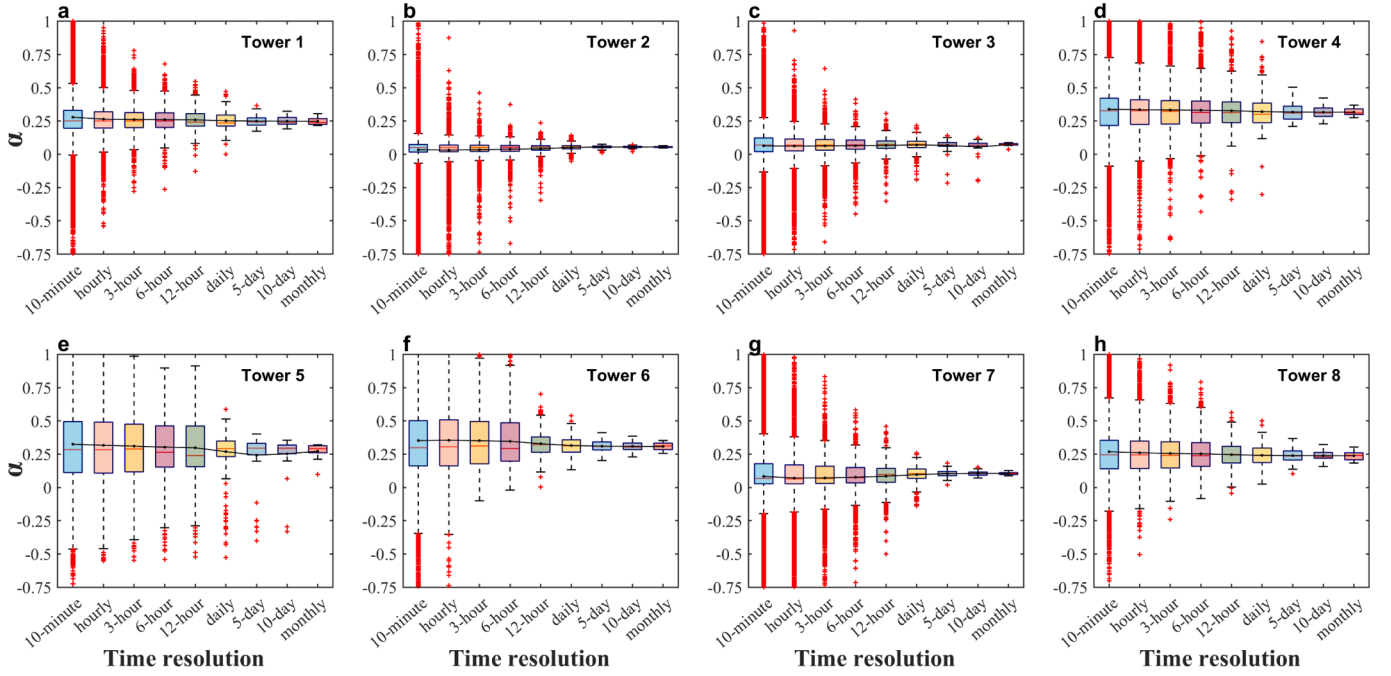


Fig. 2. Effect of time resolution on the variation of wind power law exponent α . Each box shows all fitted α value in one year. The black dotted line represents the average values. The red plus sign indicates outliers. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

atmosphere is in stable conditions (e.g., plain sites), we and other studies have shown higher mean α values than those under more unstable conditions (e.g., mountain sites). This is because the vertical convective activity is weaker at this time, making the wind speed at high altitudes more different than that near surface [21,45]. In addition, under stable atmospheric conditions in plains, some studies have shown that the daily variation of α will be larger, which corresponds to the larger variation of Tower 5&6 in our study [16,45]. These affirm the accuracy and reliability of our calculated α values. It is worth noting that, although our results have been calculated using wind speed at as many heights as possible, α calculation also varies slightly depending on the heights at which wind speeds are used, as Gualtieri [20] concluded.

3.2. Weibull distribution variation

The Weibull distribution of wind speed exhibits a gradually weakened left-skewness and an increasing peak value as the time resolution of

the data coarsens (Fig. 3a). This trend is particularly pronounced at the resolution coarser than daily. The rationale behind this observation lies in the continuous averaging of high-frequency wind speed data to coarsen resolution, which results in the elimination of extreme values. Consequently, the probability density distribution tends to converge toward a normal distribution. In Fig. 3b, the coefficient of determination R^2 for Weibull regression reveals that from the original data to daily resolution, the R^2 consistently exceeds 0.8, which corresponds to the extremely high R^2 obtained in related study using 10-minute resolution data and maximum likelihood method [22]. The relatively poor R^2 of Tower 2 is related to the more frequent occurrence of extreme wind speeds in its desert terrain, which makes it less consistent with the Weibull distribution. However, four towers exhibit R^2 values below 0.6 at the 5-day resolution, and seven towers have R^2 values below 0.6 at the monthly resolution. This underscores the coarser time resolution may lead to decreased consistency between wind speed data and the Weibull distribution. Consequently, studies employing daily wind speed data

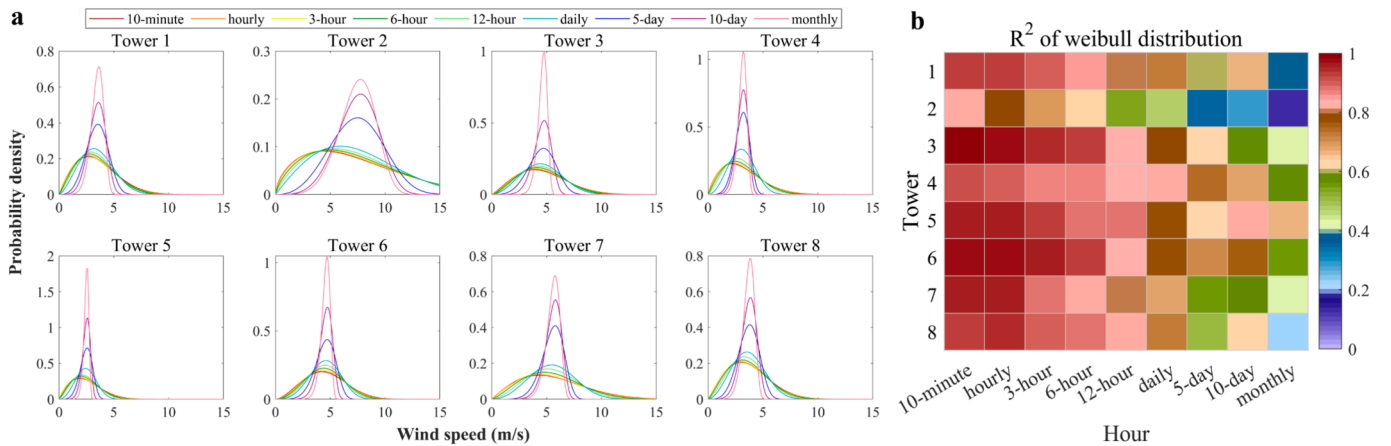


Fig. 3. Effect of time resolution on the variation of Weibull distribution. Each tower contains all valid data for one year. (a) Weibull probability density curve obtained from hub-height wind speed data at each resolution. (b) R^2 is the coefficient of determination for each fitted curve, representing the good-of-fit.

and requiring wind speed distribution should contemplate other distributions such as Gamma, Exponential, Lognormal, or composite models [15,46,47]. Moreover, it implies that the probability density distribution derived from daily and coarser data in Fig. 3a may not directly hold practical significance.

3.3. WPD errors due to coarse time resolution

The theoretical WPD is illustrated in Fig. 4, assuming that the data at various time resolutions are aggregated from the original wind speed data. Across all eight towers, a decrease of around 33–59 % in theoretical WPD is observed as the time resolution changes from 10-minute to monthly. This reduction follows a statistical principle where WPD is proportional to the cube of wind speed and extreme values are eliminated due to data averaging. In theory, coarsening the time resolution of wind speed data may lead to a decrease in the assessable wind energy, but the deviation in results varies when employing commonly used WPD calculation procedures. We also find that during some resolutions, the wind speed at higher heights is smaller than the wind speed at lower heights (70–90 m in Tower 3 and 120–140 m in Tower 5), which may be due to a combination of reversal of the wind speed profile caused by instantaneous extreme wind speeds at some heights in complex terrains and averaging of wind speed data to a coarse resolution [48,49].

We conduct analysis quantifying the WPD errors of wind speed extrapolation method and Weibull downscaling method at various time resolutions relative to hub-height theoretical WPD calculations. In addition to the original resolution, we evaluate commonly occurring hourly, daily, and monthly resolutions. As depicted in Fig. 5a, we utilize hub-height wind speed data as well as extrapolate from 10 m wind speed to hub-height, the average errors for the four resolutions in the upper figure are as follows: -0.02% (10-minute), -1.75% (hourly), -25.88% (daily), and -50.07% (monthly). Conversely, the average errors in the lower figure are: $+5.03\%$ (10-minute), $+0.27\%$ (hourly), -22.57% (daily), and -44.28% (monthly). This suggests that the fluctuation of the α value in the power law minimally affects WPD results.

Interestingly, despite the α value's instability in high-frequency data, hourly calculations may yield better results than those at 10-minute intervals. Time resolution significantly impacts Weibull WPD, notably underestimating values in daily and monthly data, which Shin et al. [28] and Gil Ruiz et al. [8] introduce even larger variations. Fig. 5b illustrates a strong correlation between Weibull WPD and theoretical WPD across eight towers at respective resolutions, with R^2 values consistently above 0.98. This means that for the same time resolution, although Fig. 3b indicates poor fit for the Weibull distribution at coarse resolutions, its effect on Weibull WPD errors compared with theoretical WPD is negligible. It suggests that the underestimation of Weibull WPD at coarse resolutions in Fig. 5a primarily stems from changes in the shape of the probability density distribution (Weibull factors) rather than the good-of-fit of the distribution.

3.4. Fine resolution Weibull factors and WPD obtained by regression

To gain deeper insights into the influence of the shape of Weibull distribution on WPD, we examine the k and c factors across different time resolutions. As illustrated in Fig. 6a, time resolution minimally affects the c factors, which remain essentially unchanged. The distribution of c factor across the eight anemometer towers exhibits considerable spatial variation, ranging from 2.55 to 8.43. This variability primarily impacts the wind speed corresponding to the peak of the Weibull distribution and mean value [50]. In Fig. 6b, the k factors derived from 10-minute high-frequency data for all towers cluster around 2, and the value is consistent with previous studies based on minute-level data [51–53]. When $k = 2$, it is also considered as the Rayleigh distribution, a fairly typical value for many locations [42]. We find that with coarser resolutions, the k factor gradually increases, leading to a more concentrated probability density distribution. For instance, k factor of Tower 5 escalates from 1.88 (10-minute) to 13.07 (monthly). Some previous studies have reported k factor of daily-level may be larger than hourly-level, but we obtain k factors with a wider variety of time resolution [7,27,54]. We indicate that errors in Weibull

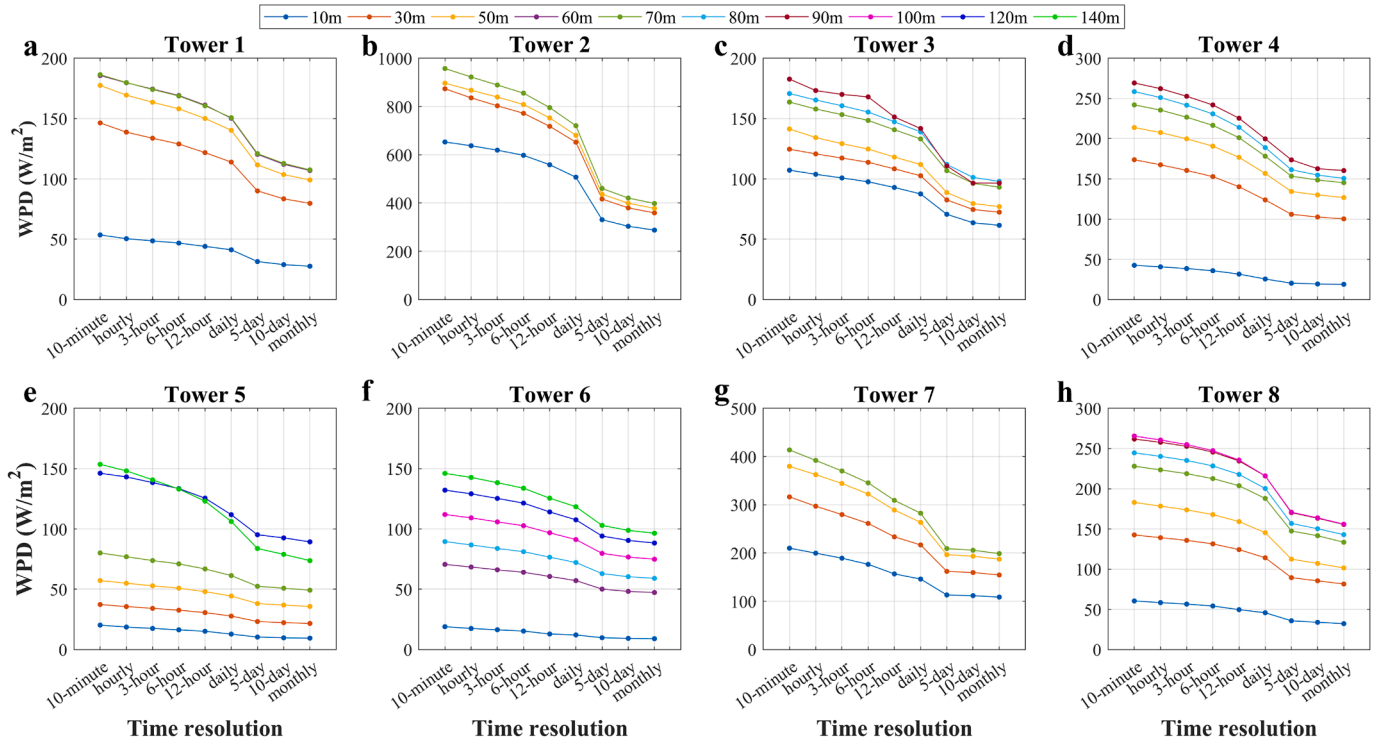


Fig. 4. Theoretical WPD from wind speed with different time resolution in eight towers. Theoretical wind power density for all heights of eight towers are shown here.

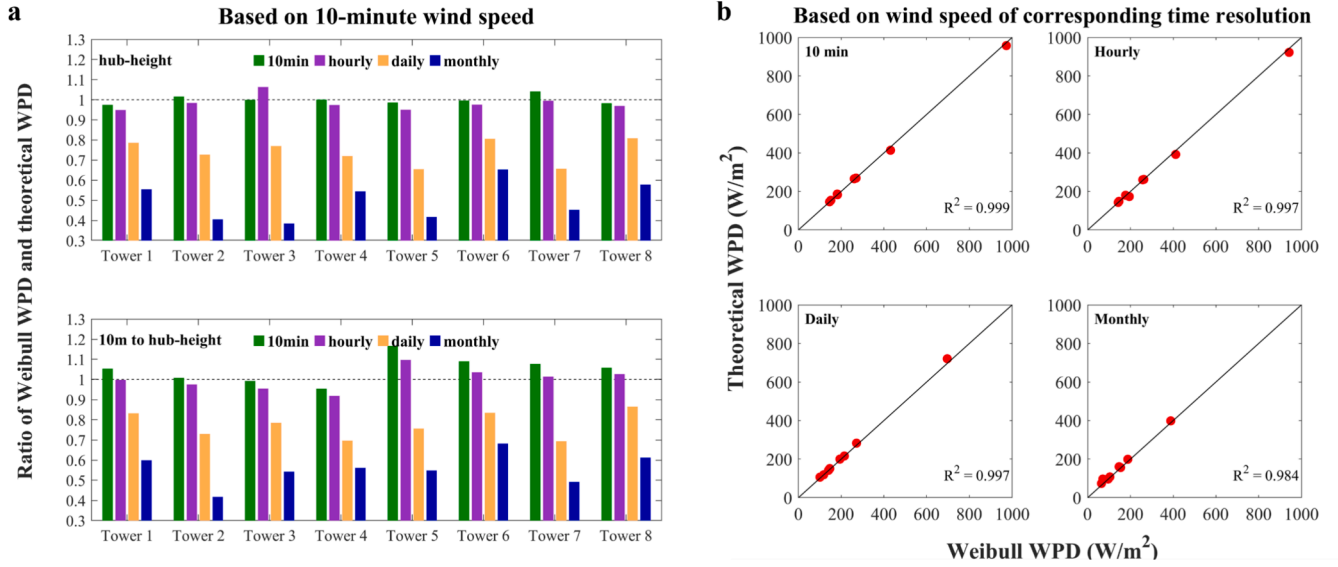


Fig. 5. Effect of time resolution of wind speed and wind speed extrapolation on wind power density errors. (a) The denominator of the value corresponding to each bar is the theoretical WPD calculated from the 10-minute raw data of the hub-heights. The numerator in the upper figure is the WPD calculated using the Weibull distribution for data at 4 common resolutions of the hub-heights. The numerator in the lower figure is first extrapolate the wind speed at a height of 10 m at different resolutions to the hub-heights using power law, and then calculate WPD using the Weibull distribution. The closer the value is to 1, the closer the result is to the real WPD. (b) Each figure uses the theoretical method and the Weibull method to calculate the WPD at the same time resolution. R^2 values calculate the coefficient of determination of the 1:1 line.

WPD estimation at coarse resolutions primarily stem from significant deviations in the k factor. To mitigate errors in WPD calculation from coarse-resolution data, further investigation into the variation of k factor with time resolution is warranted.

We propose a convenient method for regressing the k factor at coarse resolution to that at fine resolution. Given the acceptable accuracy of hourly Weibull WPD but significant errors in daily Weibull WPD (refer to section 3.3), our aim is to derive hourly k factors from daily data to mitigate WPD errors. Leveraging the insights from Fig. 6b, we posit that a straightforward formula can describe the relationship between time resolution and the k factor. As depicted in Fig. 3b, Weibull's fitting efficacy diminishes beyond the 5-day resolution. Through Section 2.5, the regressive k factors, when compared with daily data, align more closely with hourly k factors. In Fig. 6c, employing hourly k factors obtained through regression and assuming a constant c factor, we compute the adjusted Weibull WPD. These results reveal that WPD calculated using extrapolated hourly k factors from daily data closely approximates WPD derived directly at hourly resolution, demonstrating the significance of regressing daily k factors to hourly equivalents via a simple formula for improved WPD accuracy. Notably, Tower 2's regression results are poorer than daily results, possibly due to its poorer good-of-fit of Weibull distribution of daily data. This unique scenario leads to less pronounced changes in wind speed probability density shape when averaged to coarser resolutions, resulting in a smaller k factor variation, which we consider an exceptional case. Generally, we posit that this simple method can enhance the accuracy of wind energy quantification for studies that rely solely on daily wind speed data by regressing hourly k factors using the power or exponential formula.

4. Conclusions

This study explores the effect of time resolution of wind speed data on WPD. We use 9 resolutions of wind speed data at hub-heights to analyze the variations in the results caused by wind speed extrapolation and Weibull regression processes respectively. Then, the two processes are combined to quantify the WPD errors. We propose a method that can reduce the WPD errors using coarse resolution data. The study offers essential insights for selecting wind speed data and achieving accurate

wind energy assessments.

Our main findings are summarized as follows:

- 1) Reducing the time resolution of wind speed data leads to a systematic underestimation of wind power density.
- 2) Our analysis reveals that hourly wind speed data yield sufficiently accurate WPD estimations (errors $< \pm 2\%$), whereas daily data errors can reach approximately 25%.
- 3) We explain the core reasons and identify solutions to the errors. The power law exponent α , critical for wind speed extrapolation, stabilizes at coarser resolutions, thereby reducing fluctuations and potential transient conditions found in finer-resolution data. Additionally, the Weibull distribution shape concentrates more with coarser resolutions, yet Weibull WPD closely approximates theoretical values when assessed at uniform resolutions, suggesting deviations in WPD are mainly due to the increased shape factor (k) of the Weibull distribution.
- 4) We propose an innovative extrapolation method for the k factor from coarse to fine resolutions through simple nonlinear regression, effectively reducing WPD deviations.

The limitations of this study can be summarized in two points: 1) The towers used have regional constraints, and the characteristics observed regarding time resolution may not be fully applicable to other regions or terrains. 2) To obtain hub-height wind speed, the hub-heights of the eight towers differ, which may influence the results due to the height variations.

Future research can explore the global application of our results to accurately assess global wind energy potential. This will enhance the reliability of global wind energy assessments, particularly when dealing with limited time resolution in wind speed data.

CRedit authorship contribution statement

Xinrong Yang: Writing – original draft, Validation, Methodology, Investigation, Formal analysis. **Yiheng Tao:** Writing – review & editing. **Yubin Jin:** Writing – review & editing. **Bin Ye:** Writing – review & editing. **Fan Ye:** Writing – review & editing, Data curation. **Weili Duan:**

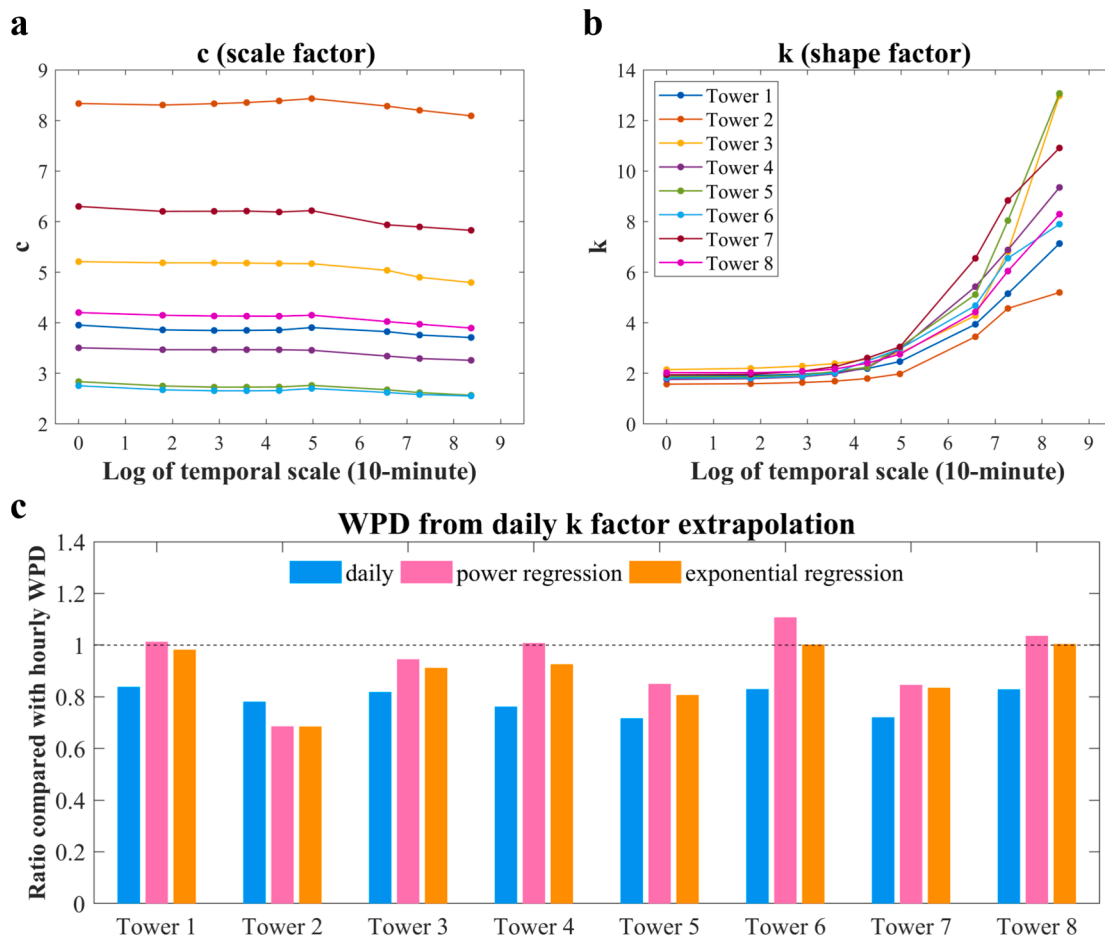


Fig. 6. Effect of time resolution on the two Weibull factors and the errors in WPD after extrapolating the daily k factor to the hourly k factor. (a) c factor of hub-height wind speed under different resolution data from eight towers. Here the temporal scale is in 10-minute units, and the x-axis is the logarithm of the temporal scale. (b) k factor of hub-height wind speed under different resolution data from eight towers. (c) The denominator of the y-axis ratio value is the Weibull WPD calculated from the hourly data of each tower. Each tower compares the Weibull WPD calculated directly using daily data after using the power and exponential formula to extrapolate the daily k factor into the hourly k factor, respectively. The closer the value is to 1, the closer the result is to the hourly WPD.

Writing – review & editing. **Rongrong Xu:** Writing – review & editing. **Zhenzhong Zeng:** Writing – review & editing, Supervision, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.ecmx.2024.100753>.

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Data availability

Data will be made available on request.

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